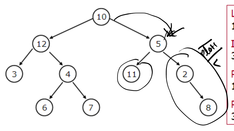
Lec-32

27/06/26 10:12 AM

in -> Left, Root, Right

level -> Root, Left, Right

preorder = [3,9,20,15,7], inorder = [9,3,15,20,7]

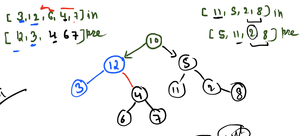


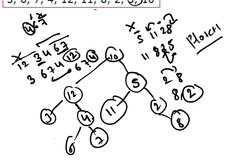
Levelorder tree traversal
10, 12, 5, 3, 4, 11, 2, 6, 7, 8

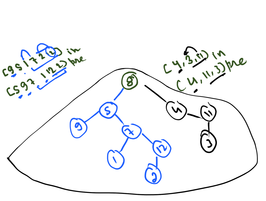
Inorder tree traversal
3, 12, 6, 4, 7, 10, 11, 5, 2, 8

Preorder tree traversal
10, 12, 3, 4, 6, 7, 5, 11, 2, 8

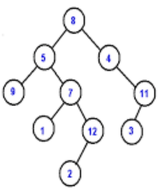
Postorder tree traversal
3, 6, 7, 4, 12, 11, 8, 2, 5, 10



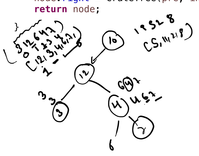


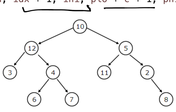


PreOrder - 8, 5, 9, 7, 1, 12, 3, 4, 11, 2
InOrder - 9, 5, 1, 7, 2, 12, 8, 4, 3, 11
PostOrder - 9, 1, 2, 12, 7, 5, 3, 11, 4, 8
LevelOrder - 8, 5, 4, 9, 7, 11, 1, 12, 3, 2



```
public TreeNode CrateTree(int[] pre, int[] in, int ilo, int ihi, int plo, int phi) {
        if (ilo > ihi || plo > phi) {
            return null;
        }
        TreeNode node = new TreeNode(pre[plo]);
        int idx = search(in, ilo, ihi, pre[plo]);
        int c = idx - ilo;
        node.left = CrateTree(pre, in, ilo, idx - 1, plo + 1, plo + c);
        node.right = CrateTree(pre, in, idx + 1, ihi, plo + c + 1, phi);
        return node;
    }
```



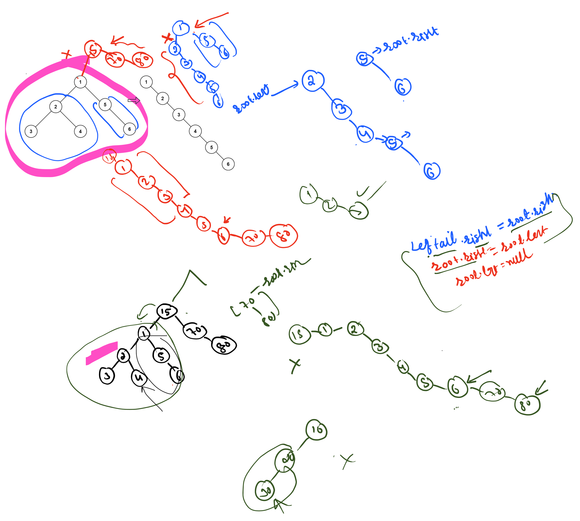
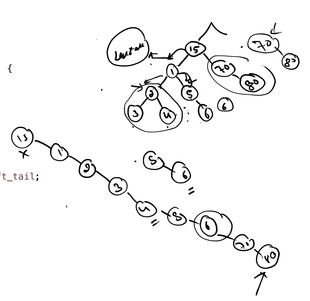
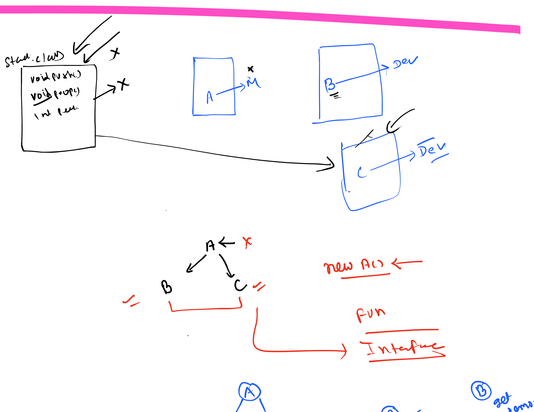
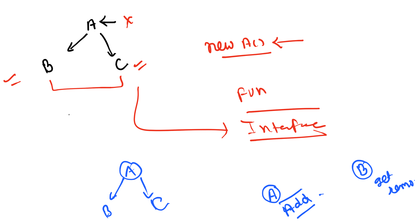


Levelorder tree traversal
10, 12, 5, 3, 4, 11, 2, 6, 7, 8

Inorder tree traversal
3, 12, 6, 4, 7, 10, 11, 5, 2, 8

Preorder tree traversal
10, 12, 3, 4, 6, 7, 5, 11, 2, 8

Postorder tree traversal
3, 6, 7, 4, 12, 11, 8, 2, 5, 10



```
public TreeNode MakeLL(TreeNode root) {
        if (root == null) {
            return null;
        }
        if (root.left == null && root.right == null) {
            return root;
        }
        TreeNode left_tail = MakeLL(root.left);
        TreeNode right_tail = MakeLL(root.right);
        if (left_tail != null) {
            left_tail.right = root.right;
            root.right = root.left;
            root.left = null;
        }
        return right_tail != null ? right_tail : left_tail;
    }
```